MGL High-Precision Compute Engine

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Masao Gate Labs development report. Validation results are from CPU release v1.1.0-dev.2. Timing measurements are from v1.1.0-dev.1.

Core, adapter, and extensions

Function	Demonstrated capability
Core	80-significant-digit arithmetic and verification. All 200 requests matched their v1.0.0 results.
Adapter	Local research tools can invoke the 80-digit arithmetic and the added mathematical functions.
Extensions	16, 32, 80, 160, or 320 significant digits. Matrix products, determinants, inverses, and linear solves up to 8 by 8.
Coefficient checks	Two computation paths matched every coefficient in 15 HJO finite-polynomial cases.

Validation on Linux

On Linux x86-64, 66 self-test methods and 43 integration cases passed. Regression checks covered 61 rational matrix products. The Windows x64 record covers 80-digit arithmetic in v1.0.0. Windows execution of the development release with matrix functions remains unverified.

Timing for eight requests

The same eight requests ran on the same Linux system. Each mode ran 30 times in alternating order. Timing included startup, computation, verification, saving, and audit.

Mode	Median	95th percentile
Eight separate launches	1.498 s	1.610 s
One batch of eight	0.312 s	0.343 s

The batch median was about 21% of the separate-launch median.

NVIDIA cuBLAS and Ozaki Scheme II

CUDA Toolkit 13.4 uses Ozaki Scheme II for cuBLAS FP64 matrix multiplication when it improves performance. NVIDIA release notes

Aspect	MGL engine	NVIDIA cuBLAS
Workload	CPU multiprecision arithmetic and verification, including small matrices	GPU matrix multiplication with FP64 emulation
Metric	0.312 s median for eight requests, including verification and saving	Published real DGEMM throughput of up to 175 TFLOPS on B200 and 45 TFLOPS on RTX PRO 6000 Blackwell Server Edition

A speed comparison requires matched inputs, precision, and timing boundaries.

References Ozaki Scheme II paper (arXiv v4) · RIKEN R-CCS GEMMul8 implementation · Development and validation reports

Pinned web record